Lecture 11 Notes: Bonds and Their Valuation

Corporate Finance
Chapter 7

I. Assignment & Review of Forward Rates

- The instructor confirms that the previous topic on forward rates is understood.
- **Assignment:** Solve the following problems and upload them to Google Classroom.

Questions: 6-2, 6-5, 6-7, 6-11, 6-14

II. Introduction to Bonds (Chapter 7)

Companies have two major sources of funding: **Debt** and **Equity**. A bond is a major component of the debt source.

What is a Bond?

A bond is a **medium- or long-term debt instrument** used by corporations or governments to raise funds. It is a contractual agreement where the borrower agrees to:

1. Return the **principal amount** at a specified future date (maturity).
2. Pay interest at a **stated rate** on specific dates to the bondholders.

Bondholders are creditors, not owners. They do not have voting rights. Their return comes from the cash flows generated by the projects financed with their funds.

Key Features

- **Maturity Date:** The specific date when the principal must be repaid.
- **Bond Indenture:** The legal document containing all terms and conditions of the bond issue.
- **Par Value / Face Value:** The principal amount (typically $1,000 in the U.S.).
- **Liquidity:** Bonds are traded in the Over-the-Counter (OTC) market.
- **Pricing:**
  - **Par:** Market price equals face value ($1,000).

  - **Discount:** Market price is less than face value.
  - **Premium:** Market price is greater than face value.

Coupon Rate vs. Yield to Maturity (YTM)

- **Coupon Interest Rate:** The stated interest rate on the bond.
- **Coupon Payment:** The dollar amount of interest.

$$\text{Coupon Payment} = \text{Par Value} \times \text{Coupon Rate}$$

- **Yield to Maturity (YTM):** The "promised yield." It is the rate of return an investor expects to earn if the bond is held until maturity.

Bond Valuation Variables

There are five key variables in bond valuation. If four are known, the fifth can be calculated:

Face Value (FV)   Present Value (PV / Market Price)   Payment (PMT / Coupon)   Interest Rate (Y

III. Bond Valuation Theory

The value of any financial asset is the **present value of its future cash flows**. For a bond, these cash flows are:

1. The periodic interest payments (an annuity).
2. The face value paid at maturity (a single sum).

$$\text{Bond Value} = PV_{\text{Annuity}} + PV_{\text{Lump Sum}}$$

Example 1: Coupon Rate

A 10-year bond with a 10% coupon rate and a 10% YTM.

$$\text{Value} = \$1,000$$

Conclusion: When the coupon rate equals the YTM, the bond trades at **par**.

Example 2: Coupon Rate $>$ YTM (Premium Bond)

A 10-year bond with a **13% coupon rate** and a **10% YTM**.

- **Annual Coupon Payment:** $13\% \times \$1,000 = \$130$
- **Number of Periods (n):** 10 years
- **Discount Rate (r):** 10% (the YTM)

The market price is calculated as:

$$\begin{aligned}
\text{PV of Annuity} &= \$130 \times \left[\frac{1-(1.10)^{-10}}{0.10}\right] = \$130 \times 6.144567 = \$798.81 \\
\text{PV of Face Value} &= \$1,000 \times (1.10)^{-10} = \$1,000 \times 0.385543 = \$385.54 \\
\text{Bond Value (Price)} &= \$798.81 + \$385.54 = \$1,184.35
\end{aligned}$$

Conclusion: Since the coupon rate (13%) is higher than the YTM (10%), the bond trades at a **premium**.

Example 3: Coupon Rate $<$ YTM (Discount Bond)

A 10-year bond with a **7% coupon rate** and a **10% YTM**.

- **Annual Coupon Payment:** $7\% \times \$1,000 = \$70$
- **Number of Periods (n):** 10 years
- **Discount Rate (r):** 10% (the YTM)

The market price is calculated as:

$$\begin{aligned}
\text{PV of Annuity} &= \$70 \times \left[\frac{1-(1.10)^{-10}}{0.10}\right] = \$70 \times 6.144567 = \$430.13 \\
\text{PV of Face Value} &= \$1,000 \times (1.10)^{-10} = \$1,000 \times 0.385543 = \$385.54 \\
\text{Bond Value (Price)} &= \$430.13 + \$385.54 = \$815.67
\end{aligned}$$

Conclusion: Since the coupon rate (7%) is lower than the YTM (10%), the bond trades at a **discount**.

IV. Key Relationships

- **Coupon Rate = YTM:** Bond sells at **Par**.
- **Coupon Rate ¿ YTM:** Bond sells at a **Premium**.
- **Coupon Rate ¡ YTM:** Bond sells at a **Discount**.

V. Semi-Annual Bonds

Most bonds pay interest semi-annually. For these bonds, all variables must be adjusted:

- **Periods (n):** Years $\times$ 2
- **Semi-Annual Coupon Payment:** $\frac{\text{Annual Coupon}}{2}$
- **Semi-Annual Discount Rate (YTM):** $\frac{\text{Annual YTM}}{2}$

These adjustments will be covered in the next lecture.

VI. Logistics

- **Exams:** Will be physical.
- **Assignment Deadline:** Wednesday.
- **Attendance:** 36 students present (excluding instructor).